

HSC Extension 2 Mathematics

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Further Integration



Which of the following is false?

A) $\int_{-3}^3 x^3 e^{-x^2} dx = 0$

B) $\int_{-4}^4 \frac{x^2}{x^2 + 4} dx = 2 \int_0^4 \frac{x^2}{x^2 + 4} dx$

C) $\int_0^{\pi} \sin^4 \theta d\theta > \int_0^{\pi} \sin 4\theta d\theta$

D) $\int_0^1 x^4 dx < \int_0^1 x^5 dx$



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Find

(i) $\int \frac{dx}{x(\ln x)}$

2

(ii) $\int \frac{1}{x^2 + 6x + 13} dx$

2



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Further Integration



a) Find $\int \frac{dx}{x\sqrt{x^2 - 4}}$ using the substitution $u = \sqrt{x^2 - 4}$ 2

b) Evaluate $\int x^2 \sin x dx$ 2



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(i) If $I_n = \int_0^{\frac{\pi}{4}} \sec^n \theta d\theta$ show that

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$$I_n = \frac{1}{n-1} [(n-2)I_{n-2} + (\sqrt{2})^{n-2}]$$

for $n \geq 2$

(ii) Evaluate $\int_0^{\frac{\pi}{4}} \sec^4 \theta d\theta$

2



Further Integration Solution

$$(i) \int \frac{dx}{x(\ln x)}$$

$$\text{Let } u = \ln x$$

$$\therefore \frac{du}{dx} = \frac{1}{x}$$

$$\therefore du = \frac{dx}{x}$$

$$\int \frac{du}{u} \quad (1)$$

$$= \log_e u$$

$$= \log_e (\log_e x) + C \quad (2)$$

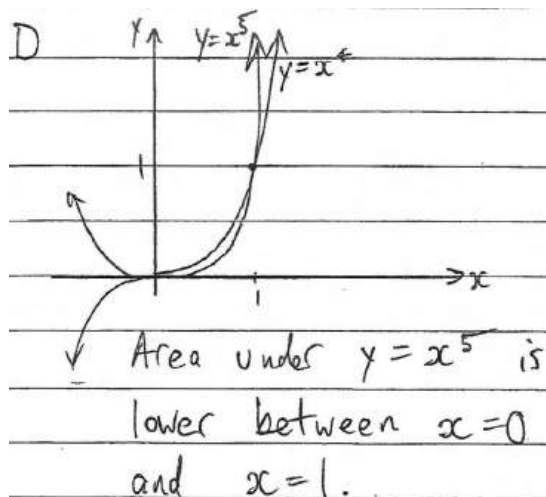
$$(ii) \int \frac{1}{x^2 + 6x + 13} dx$$

$$\int \frac{1}{(x+3)^2 + 2^2} dx \quad (1)$$

$$= \frac{1}{2} \tan^{-1} \left(\frac{x+3}{2} \right) + C \quad (2)$$



Further Integration Solution



Further Integration Solution

$$I_n = \int_0^{\pi/4} \sec^n \theta d\theta$$

$$= \int_0^{\pi/4} \sec^2 \theta \sec^{n-2} \theta d\theta$$

$$= \int_0^{\pi/4} \frac{1}{\cos^2 \theta} (\tan \theta) \sec^{n-2} \theta d\theta \quad (1)$$

$$= [\tan \theta \sec^{n-2} \theta]_0^{\pi/4} - \int_0^{\pi/4} \tan \theta \times (n-2) \sec^{n-3} \theta \times \sec \theta \tan \theta d\theta$$

$$= (\sqrt{2})^{n-2} - \int_0^{\pi/4} (n-2) \tan^2 \theta \sec^{n-2} \theta d\theta$$

$$= (\sqrt{2})^{n-2} - (n-2) \int_0^{\pi/4} (\sec^2 \theta - 1) \sec^{n-2} \theta d\theta$$

$$= (\sqrt{2})^{n-2} - (n-2) \int_0^{\pi/4} \sec^n \theta d\theta + (n-2) \int_0^{\pi/4} \sec^{n-2} \theta d\theta$$

$$I_n = (\sqrt{2})^{n-2} - (n-2) I_n + (n-2) I_{n-2} \quad (2)$$

$$I_n + (n-2) I_n = (\sqrt{2})^{n-2} + (n-2) I_{n-2}$$

$$(n-1) I_n = (n-2) I_{n-2} + (\sqrt{2})^{n-2}$$

$$I_n = \frac{1}{n-1} [(n-2) I_{n-2} + (\sqrt{2})^{n-2}] \quad (3)$$

$$(iv) I_4 = \frac{2}{3} I_2 + \frac{1}{3} (\sqrt{2})^2 \quad (1)$$

$$= \frac{2}{3} (0 + 1) + \frac{2}{3}$$

$$I_4 = \frac{4}{3} \quad (2)$$



Further Integration Solution

$$a) \int \frac{dx}{x \sqrt{x^2 - 4}}$$

$$\text{Let } u = \sqrt{x^2 - 4}$$

$$du = \frac{2x}{2\sqrt{x^2 - 4}} dx$$

$$\int \frac{1}{x} \times \frac{du}{u}$$

$$\frac{du}{x} = \frac{dx}{\sqrt{x^2 - 4}}$$

$$\int \frac{du}{u^2}$$

$$\int \frac{du}{u^2 + 4} \quad (1) \quad \text{Since } u^2 = x^2 - 4$$

$$= \frac{1}{2} \tan^{-1} \frac{u}{2} + C$$

$$= \frac{1}{2} \tan^{-1} \frac{\sqrt{x^2 - 4}}{2} + C \quad (2)$$

$$b) \int \frac{1}{x^2} \sin x dx$$

$$\int x^2 \frac{1}{x^2} (-\cos x) dx$$

$$= -x^2 \cos x - \int 2x \times -\cos x dx$$

$$= -x^2 \cos x + 2 \int x \frac{1}{x} (\sin x) dx \quad (1)$$

$$= -x^2 \cos x + 2 [x \sin x - \int \sin x dx]$$

$$= -x^2 \cos x + 2x \sin x + 2 \cos x + C \quad (2)$$



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Further Integration



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- (i) Find real numbers a, b and c such that

$$\frac{1}{x^2(x-1)} = \frac{a}{x} + \frac{b}{x^2} + \frac{c}{x-1}$$

- (ii) Hence find $\int \frac{1}{x^2(x-1)} dx$ in simplest form.

2



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Further Integration



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- (i) Using a suitable substitution, show that

$$\int_0^a f(x) dx = \int_0^a f(a-x) dx$$

- (ii) A function $f(x)$ has the property that $f(x) + f(a-x) = f(a)$

2

Using part (i), or otherwise, show that

$$\int_0^a f(x) dx = \frac{a}{2} f(a)$$



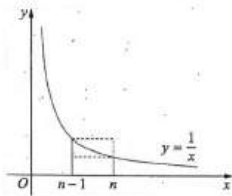
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Further Integration



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Let n be a positive integer greater than 1. The area of the region under the curve

$y = \frac{1}{x}$ from $x = n-1$ to $x = n$ is between the areas of two rectangles, as shown in the diagram.

Show that $e^{-n} < \left(1 - \frac{1}{n}\right)^n < e^{-1}$



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Further Integration



- (i) Determine constants A and B such that $\frac{5}{(x-1)(3x+2)} = \frac{A}{x-1} + \frac{B}{3x+2}$. 1

- (ii) Hence find the particular solution of the differential equation 3

$$v \frac{dv}{dx} = \frac{5v^2}{(x-1)(3x+2)} \quad \text{for } x > 1,$$

for which $v(2) = 8$. Give your answer in the form $v = f(x)$.



Further Integration Solution

$$i) \int_0^a f(x) dx = \int_0^a f(a-x) dx$$

In RHS, let $u = a-x$ when $x=0$, $u=a$
 $du = -dx$ when $x=a$, $u=0$ ①

$$\therefore \text{RHS becomes } \int_a^0 f(u) (-du) \\ = \int_0^a f(u) du \quad ② \\ = \text{LHS (different variable irrelevant)}$$

$$ii) \int_0^a f(x) + f(a-x) dx = \int_0^a f(a) dx \\ \int_0^a f(x) dx + \int_0^a f(a-x) dx = \int_0^a f(a) dx \\ 2 \int_0^a f(x) dx = [f(a)x]_0^a \quad ①$$

$$2 \int_0^a f(x) dx = a f(a)$$

$$\therefore \int_0^a f(x) dx = \frac{a}{2} f(a)$$



Further Integration Solution

$$i) 1 = ax(x-1) + b(x-1) + cx^2$$

Let $x=1 \therefore 1 = c$ ① for
 $x=0 \therefore 1 = -b$ ^{one of those}

$$\text{Let } x=2 \therefore 1 = 2a + b + 4c \\ 1 = 2a - 1 + 4$$

$$\text{So } a = -1, b = -1, c = 1 \quad ②$$

$$ii) \int \frac{-1}{2x} - \frac{1}{x^2} + \frac{1}{x-1} dx \\ = -\log_e |x| + x^{-1} + \log_e |x-1| \quad ① \\ = \log_e \left| \frac{x-1}{2x} \right| + \frac{1}{x} + C \quad ②$$



Further Integration Solution

$$i) \frac{5}{(x-1)(3x+2)} = \frac{A}{x-1} + \frac{B}{3x+2}$$

$$A = \frac{5}{3 \cdot 1 + 2} \quad (\text{By Cover-Up Rule})$$

$$B = \frac{5}{(-\frac{2}{3}-1)} = \frac{1}{x-1} - \frac{3}{3x+2} \quad \checkmark$$

$$ii) v \frac{dv}{dx} = \frac{5x^2}{(x-1)(3x+2)} \quad \text{Constant solution } v=0$$

$$\int \frac{1}{v} dv = \int \frac{5}{(x-1)(3x+2)} dx \quad \checkmark$$

$$\ln |v| = \int \frac{1}{x-1} - \frac{3}{3x+2} dx \quad \checkmark$$

$$\ln |v| = \ln(x-1) - \ln(3x+2) + C \quad (x > 1)$$

$$\ln \left| \frac{v(3x+2)}{x-1} \right| = C \quad (C \in \mathbb{R})$$

$$\frac{v(3x+2)}{x-1} = A \quad (A \in \mathbb{R})$$

$$v = \frac{A(x-1)}{3x+2} \quad \checkmark$$

Initial conditions

$$v(2) = 8 \quad 8 = \frac{A(2-1)}{3 \cdot 2 + 2}$$

$$8 = \frac{A}{8}$$

$$A = 64$$

$$v = \frac{64(x-1)}{3x+2} \quad \checkmark$$

Do not penalise incorrect treatment of absolute values and constant



Further Integration Solution

$$\text{Area of lower rectangle} = \frac{1}{n}$$

$$\text{Area of upper rectangle} = \frac{1}{n-1}$$

$$\therefore \frac{1}{n} < \int_{n-1}^n \frac{dx}{x} < \frac{1}{n-1}$$

$$\frac{1}{n} < [\ln x]_{n-1}^n < \frac{1}{n-1}$$

$$\frac{1}{n} < \ln \frac{n}{n-1} < \frac{1}{n-1}$$

$$e^{\frac{1}{n}} < \frac{n}{n-1} < e^{\frac{1}{n-1}} \quad ①$$

$$e < \left(\frac{n}{n-1} \right)^n < e^{\frac{n}{n-1}}$$

$$\frac{1}{e} > \left(\frac{n-1}{n} \right)^n > \frac{1}{e^{\frac{n}{n-1}}}$$

$$e^{\frac{1}{n-1}} < \left(\frac{n-1}{n} \right)^n < \frac{1}{e}$$

$$e^{\frac{n}{n-1}} < \left(1 - \frac{1}{n} \right)^n < e^{-1} \quad \text{as req'd.} \quad ②$$



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Further Integration



Use a suitable substitution to evaluate $\int_1^4 \frac{e^{\sqrt{x}}}{\sqrt{x}} dx$.

2



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Further Integration



Let $I_n = \int_0^{\frac{\pi}{4}} \sec^n x dx$.

(i) Use integration by parts to show that for $n > 2$,

3

$$I_n = \frac{1}{n-1} \left((\sqrt{2})^{n-2} + (n-2)I_{n-2} \right).$$

(ii) Show that $\frac{d}{dx} (\ln(\sec x + \tan x)) = \sec x$.

1

(iii) Hence show that $\int_0^{\frac{\pi}{4}} \sec x dx = \ln(\sqrt{2} + 1)$ and find a simplified expression for I_5 .

3



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Further Integration



Consider the improper integral $I = \int_1^{\infty} \frac{1}{1+x^3} dx$.

(i) Use the substitution $u = \frac{1}{x}$ to show that $I = \int_0^1 \frac{u}{1+u^3} du$.

2

(ii) Hence or otherwise evaluate $\int_0^{\infty} \frac{1}{1+x^3} dx$.

3



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Further Integration



Evaluate $\int_0^{\frac{\pi}{4}} \tan^3 x \sec^2 x dx$.

2



Further Integration Solution

$$(i) \int_0^{\pi/4} \sec^3 x \, dx = \int_0^{\pi/4} \sec x \sec^2 x \, dx$$

Let $u = \sec x$
 $u' = (\sec x) \tan x$
 $\times \sec x \tan x$
 $= (\sec x) \tan^2 x$
 $V = \sec x$
 $V = \tan x$

$$I_n = \left[\sec x \tan x \right]_0^{\pi/4} - \int_0^{\pi/4} \tan^2 x \sec x \, dx$$

✓ CONNECT INT BY PARTS

$$= \left[\left(\frac{1}{\cos x} \right) \tan x - 0 \right] - \int_0^{\pi/4} (\sec^2 x - 1) \sec x \, dx$$

$$I_n = (\sqrt{2})^{n-2} - (n-2) [I_n - I_{n-2}]$$

✓ RECURRENCE

$$I_n + (n-2)I_n = (\sqrt{2})^{n-2} + (n-2)I_{n-2}$$

$$(n-1)I_n = (\sqrt{2})^{n-2} + (n-2)I_{n-2}$$

✓ SHOW

$$I_n = \frac{1}{n-1} \left[(\sqrt{2})^{n-2} + (n-2)I_{n-2} \right] \text{ as reqd}$$

$$(ii) \frac{d}{dx} [\ln(\sec x + \tan x)] = \frac{1}{\sec x + \tan x} \times (\sec x \tan x + \sec^2 x)$$

(By Chain Rule)

$$= \frac{\sec x (\tan x + \sec x)}{\sec x + \tan x}$$

✓ SHOW

$$= \sec x$$

$$(iii) \int_0^{\pi/4} \sec x \, dx = \left[\ln(\sec x + \tan x) \right]_0^{\pi/4}$$

$$= \ln \left(\frac{1}{\cos \pi/4} + \tan \frac{\pi}{4} \right) - \ln(1+0)$$

✓ SHOW

$$= \ln(\sqrt{2}+1) \text{ as required}$$

From (ii) $I_1 = \ln(\sqrt{2}+1)$
 $I_3 = \frac{1}{2} [\sqrt{2} + I_1]$
 $I_5 = \frac{1}{4} [\sqrt{2}^3 + 3I_3]$
 $= \frac{1}{4} [\sqrt{2} + \frac{3}{2}(\sqrt{2} + I_1)]$
 $= \frac{1}{4} [\sqrt{2} + \frac{3\sqrt{2}}{2} + \frac{3}{2} \ln(\sqrt{2}+1)]$
 $= \frac{1}{8} [7\sqrt{2} + 3 \ln(\sqrt{2}+1)]$ ✓



Further Integration Solution

Let $u = \sqrt{x}$
 $du = \frac{1}{2} x^{-1/2} dx$
 $2du = \frac{dx}{\sqrt{x}}$

x	1	4
u	1	2

$$\int_1^2 2e^u du = [2e^u]_1^2$$

$$= 2(e^2 - e)$$

$$= 2e(e-1)$$
 ✓



Further Integration Solution

$$\int_0^{\pi/4} \tan^3 x \sec^2 x \, dx$$

$$= \left[\frac{\tan^4 x}{4} \right]_0^{\pi/4}$$

$$= \frac{1}{4}$$



Further Integration Solution

(i) $u = \frac{1}{x}$ $du = -\frac{1}{x^2} dx$ Lower limit $x=1$ $u=1$
 $x^3 = \frac{1}{u^3}$ $du = -u^2 dx$ Upper limit $\lim_{x \rightarrow \infty} \left(\frac{1}{x} \right) = 0$
 $\frac{1}{u^4} du = dx$

$$I = \int_{u=1}^0 \frac{-\frac{du}{u^4}}{1 + \frac{1}{u^3}} \quad (\times u^3)$$

$$= - \int_1^0 \frac{u \, du}{u^3 + 1} \quad (\times u^3) \quad \text{✓ SHOW}$$

$$= \int_0^1 \frac{u \, du}{1 + u^3} \quad (\text{Reversing limits})$$

(ii) $\int_0^{\infty} \frac{1}{1+x^3} dx = \int_0^1 \frac{1}{1+x^3} dx + \int_1^{\infty} \frac{1}{1+x^3} dx$ (Splitting interval)

$$= \int_0^1 \frac{1}{1+x^3} dx + \int_0^1 \frac{x}{1+x^3} dx \quad \text{✓ (Using ii and replacing dummy variable } u \text{ with } x)$$

$$= \int_0^1 \frac{1+x}{1+x^3} dx$$

$$= \int_0^1 \frac{1+x}{(1+x)(1-x+x^2)} dx$$

$$= \int_0^1 \frac{1}{(1-x+x^2)} dx \quad \text{✓ } a = \sqrt{3}/2$$

$$= \frac{1}{\sqrt{3}/2} \left[\tan^{-1} \left(\frac{x - \frac{1}{2}}{\sqrt{3}/2} \right) \right]_0^1$$

$$= \left[\frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{2x-1}{\sqrt{3}} \right) \right]_0^1$$

$$= \frac{2}{\sqrt{3}} \left[\tan^{-1} \left(\frac{1}{\sqrt{3}} \right) - \tan^{-1} \left(-\frac{1}{\sqrt{3}} \right) \right]$$

$$= \frac{2}{\sqrt{3}} \left[\frac{\pi}{6} - \left(-\frac{\pi}{6} \right) \right]$$

$$= \frac{2\pi}{3\sqrt{3}}$$

$$= \frac{2\sqrt{3}\pi}{9}$$
 ✓



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Further Integration



Find $I = \int e^x \sin x \, dx$.

3



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Further Integration



Use the substitution $t = \tan \frac{\theta}{2}$ to find $\int \frac{1}{1 + \sin \theta + \cos \theta} d\theta$.

3



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Further Integration



Find $\int \frac{x}{1 + \sqrt{x}} \, dx$.

3



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Further Integration



Let $I_n = \int_0^{\frac{\pi}{4}} \cos^n x \, dx$.

(i) Show that $I_n = \frac{1}{n} \left(\frac{1}{\sqrt{2}} \right)^n + \frac{n-1}{n} I_{n-2}$ for $n \geq 2$. 3

(ii) By using an appropriate trigonometric substitution, show that 2

$$\int_0^2 \frac{1}{\sqrt{(x^2 + 4)^7}} \, dx = \frac{1}{64} \int_0^{\frac{\pi}{4}} \cos^5 \theta \, d\theta.$$

(iii) Hence deduce that $\int_0^2 \frac{1}{\sqrt{(x^2 + 4)^7}} \, dx = \frac{43}{3840} \left(\frac{1}{\sqrt{2}} \right)$. 2



Further Integration Solution

$$\int \frac{1}{1 + \sin \theta + \cos \theta} d\theta \quad \text{Let } t = \tan \frac{\theta}{2}$$

$$\frac{dt}{d\theta} = \frac{1}{2} \sec^2 \frac{\theta}{2} = \frac{1}{2} (1+t^2)$$

$$d\theta = \frac{2dt}{1+t^2}$$

$$\int \frac{1}{1 + \frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2}} \cdot \frac{2dt}{1+t^2}$$

$$= \int \frac{2dt}{1+t^2+2t+1-t^2}$$

$$= \int \frac{2dt}{2+2t} dt$$

$$= \ln |1 + \tan \frac{\theta}{2}| + c \quad (\theta \neq (2n+1)\pi)$$



Further Integration Solution

$$I = \int e^x \sin x dx$$

$$u = \sin x \quad v' = e^x$$

$$u' = \cos x \quad v = e^x$$

$$I = e^x \sin x - \int e^x \cos x dx$$

$$u = \cos x \quad v' = e^x$$

$$u' = -\sin x \quad v = e^x$$

$$I = e^x \sin x - (e^x \cos x + \int e^x \sin x dx)$$

$$2I = e^x (\sin x - \cos x) + 2c$$

$$I = \frac{1}{2} e^x (\sin x - \cos x) + c$$



Further Integration Solution

$$\text{RTP } I_n = \frac{1}{n} \left(\frac{1}{\sqrt{x}} \right)^n + \frac{n-1}{n} \int \cos^{n-2} x dx$$

$$u = \cos^{n-1} x \quad v' = \cos x$$

$$u' = -(n-1) \sin x \cos^{n-2} x \quad v = \sin x$$

$$I_n = \left[\sin x \cos^{n-1} x \right]_0^{\pi/2} + (n-1) \int_0^{\pi/2} \sin^2 x \cos^{n-2} x dx$$

$$= \left(\frac{1}{\sqrt{x}} \right)^n + (n-1) \int_0^{\pi/2} (1 - \cos^2 x) \cos^{n-2} x dx = \left(\frac{1}{\sqrt{x}} \right)^n + (n-1) [I_{n-2} - I_n]$$

$$I_n (1 + n-1) = \left(\frac{1}{\sqrt{x}} \right)^n + (n-1) I_{n-2}$$

$$I_n = \frac{1}{n} \left(\frac{1}{\sqrt{x}} \right)^n + \frac{n-1}{n} I_{n-2}$$

$$(ii) \int_0^{\pi/2} \frac{1}{\sqrt{(x^2+4)^3}} dx \quad \text{Let } \theta = \tan^{-1} \left(\frac{x}{2} \right)$$

$$x = 2 \tan \theta$$

$$\frac{dx}{d\theta} = 2 \sec^2 \theta$$

$$\theta = \frac{\pi}{4} \Rightarrow x = 2, \theta = \frac{\pi}{4}$$

$$x = 0 \Rightarrow \theta = 0$$

$$\int_0^{\pi/4} \frac{2 \sec^2 \theta}{\sqrt{(4 + 4 \tan^2 \theta)^3}} d\theta$$

$$= \frac{1}{8} \int_0^{\pi/4} \frac{\sec^2 \theta}{\sec^3 \theta} d\theta \quad \sec \theta > 0 \text{ for } 0 \leq \theta \leq \frac{\pi}{4}$$

$$= \frac{1}{8} \int_0^{\pi/4} \cos \theta d\theta$$

$$(iii) \int_0^{\pi/4} \frac{1}{\sqrt{(x^2+4)^3}} dx = \frac{I_2}{64}$$

$$I_1 = \int_0^{\pi/4} \cos \theta d\theta = \sin \frac{\pi}{4} = \frac{1}{\sqrt{2}}$$

$$I_3 = \frac{1}{3} \left(\frac{1}{\sqrt{2}} \right)^3 + \frac{2}{3} \left(\frac{1}{\sqrt{2}} \right) = \frac{1}{6} \left(\frac{1}{\sqrt{2}} \right) + \frac{2}{3} \left(\frac{1}{\sqrt{2}} \right) = \frac{5}{6} \left(\frac{1}{\sqrt{2}} \right)$$

$$I_4 = \frac{1}{4} \left(\frac{1}{\sqrt{2}} \right)^4 + \frac{3}{4} \left(\frac{5}{6} \left(\frac{1}{\sqrt{2}} \right) \right) = \frac{1}{32} \left(\frac{1}{\sqrt{2}} \right) + \frac{15}{32} \left(\frac{1}{\sqrt{2}} \right) = \frac{16}{32} \left(\frac{1}{\sqrt{2}} \right)$$

$$\therefore \int_0^{\pi/4} \frac{1}{\sqrt{(x^2+4)^3}} dx = \frac{1}{64} \left(\frac{16}{32} \right) \left(\frac{1}{\sqrt{2}} \right) = \frac{1}{128} \left(\frac{1}{\sqrt{2}} \right)$$



Further Integration Solution

$$\int \frac{x}{1 + \sqrt{x}} dx \quad \text{Let } u = 1 + \sqrt{x}$$

$$\frac{du}{dx} = \frac{1}{2\sqrt{x}}$$

$$\frac{dx}{du} = 2\sqrt{x} du$$

$$dx = 2(u-1) du$$

$$x = (u-1)^2$$

$$\int \frac{(u-1)^2}{u} \cdot 2(u-1) du$$

$$= 2 \int \frac{(u-1)^3}{u} du$$

$$= 2 \int \frac{u^3 - 3u^2 + 3u - 1}{u} du$$

$$= 2 \int \left(u^2 - 3u + 3 - \frac{1}{u} \right) du$$

$$= 2 \left(\frac{u^3}{3} - \frac{3u^2}{2} + 3u + \ln|u| \right) + c$$

$$= 2 \left(\frac{(1+\sqrt{x})^3}{3} - \frac{3(1+\sqrt{x})^2}{2} + 6(1+\sqrt{x}) + \ln(1+\sqrt{x}) \right) + c$$



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Further Integration



- (i) Find the values of A , B , and C such that:

3

$$\frac{x^2 + x + 11}{(x+2)(x^2+9)} = \frac{A}{x+2} + \frac{Bx+C}{x^2+9}.$$

- (ii) Hence find $\int \frac{x^2 + x + 11}{(x+2)(x^2+9)} dx$.

2



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Further Integration



What is the value of this integral correct to 2 decimal places?

3

$$\int_{0.5}^1 \frac{x}{\sqrt{2x-x^2}} dx$$



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Further Integration



Use the substitution $t = \tan \frac{x}{2}$ to evaluate $\int_0^{\frac{\pi}{3}} \frac{1}{3 - \cos \theta} d\theta$.

4

Leave your answer correct to 2 decimal places.



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Further Integration



Use an appropriate trigonometric substitution to show that

3

$$\int \frac{x^2}{\sqrt{9-x^2}} dx = \frac{9}{2} \sin^{-1} \frac{x}{3} - \frac{x}{2} \sqrt{9-x^2} + C.$$



Further Integration Solution

$$\begin{aligned}
 \int_{\frac{1}{2}}^1 \frac{x}{\sqrt{2x-x^2}} dx &= -\int_{\frac{1}{2}}^1 \frac{-x}{\sqrt{2x-x^2}} dx \\
 &= -\int_{\frac{1}{2}}^1 \frac{1-x+1}{\sqrt{2x-x^2}} dx \\
 &= -\int_{\frac{1}{2}}^1 \left[\frac{1-x}{\sqrt{2x-x^2}} + \frac{1}{\sqrt{2x-x^2}} \right] dx \\
 &= -\frac{1}{2} \int_{\frac{1}{2}}^1 (2-2x)(2x-x^2)^{-\frac{1}{2}} dx - \int_{\frac{1}{2}}^1 \frac{1}{\sqrt{1-(x-1)^2}} dx \\
 &= -\frac{1}{2} \left[2(2x-x^2)^{\frac{1}{2}} \right]_{\frac{1}{2}}^1 - \left[\sin^{-1}(x-1) \right]_{\frac{1}{2}}^1 \\
 &= -\left[\sqrt{1} - \sqrt{\frac{3}{4}} \right] - \left[\sin^{-1} 0 - \sin^{-1}\left(-\frac{1}{2}\right) \right] \\
 &= 0.389624... \approx 0.39
 \end{aligned}$$



Further Integration Solution

$$\begin{aligned}
 \frac{x^2+x+11}{(x+2)(x^2+9)} &= \frac{A}{x+2} + \frac{Bx+C}{x^2+9} \\
 \therefore x^2+x+11 &\equiv A(x^2+9) + (Bx+C)(x+2)
 \end{aligned}$$

$$\text{Coefficients of } x^2: A+B=1$$

$$\text{Constants: } 9A+2C=11$$

$$\text{when } x=-2, \quad 4-2+11=13A$$

$$\therefore A=1$$

$$\therefore B=0$$

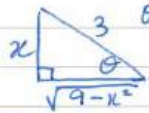
$$\therefore C=1$$

$$\begin{aligned}
 \text{ii) } \int \frac{x^2+x+11}{(x+2)(x^2+9)} dx &= \int \left(\frac{1}{x+2} + \frac{1}{x^2+9} \right) dx \\
 &= \ln|x+2| + \frac{1}{3} \tan^{-1} \frac{x}{3} + C
 \end{aligned}$$



Further Integration Solution

$$\begin{aligned}
 \text{LHS} &= \int \frac{x^2}{\sqrt{9-x^2}} dx \quad \text{Let } x=3\sin\theta \\
 &\quad dx=3\cos\theta d\theta \\
 &= \int \frac{9\sin^2\theta}{\sqrt{9-9\sin^2\theta}} \times 3\cos\theta d\theta \\
 &= \int \frac{9\sin^2\theta}{\sqrt{9(1-\sin^2\theta)}} \times 3\cos\theta d\theta \\
 &= \frac{1}{3} \int \frac{9\sin^2\theta}{\sqrt{\cos^2\theta}} \times 3\cos\theta d\theta \\
 &= 9 \int \frac{\sin^2\theta}{\cos\theta} \cdot \cos\theta d\theta \\
 &= 9 \int \sin^2\theta d\theta \\
 &= 9 \int \frac{1}{2}(1-\cos 2\theta) d\theta \\
 &= \frac{9}{2} \left[\theta - \frac{\sin 2\theta}{2} \right] + C \\
 &= \frac{9}{2} \left[\theta - \frac{2\sin\theta\cos\theta}{2} \right] + C \\
 &\quad \text{If } x=3\sin\theta \\
 &\quad \sin\theta = \frac{x}{3} \\
 &\quad \theta = \sin^{-1} \frac{x}{3} \\
 &\quad \cos\theta = \frac{\sqrt{9-x^2}}{3} \\
 &= \frac{9}{2} \left(\sin^{-1} \frac{x}{3} - \frac{x}{3} \times \frac{\sqrt{9-x^2}}{3} \right) + C \\
 &= \frac{9}{2} \sin^{-1} \frac{x}{3} - \frac{x\sqrt{9-x^2}}{2} + C = \text{RHS}
 \end{aligned}$$



Further Integration Solution

$$\begin{aligned}
 I &= \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{1}{3-\cos\theta} d\theta \quad \text{If } t = \tan \frac{\theta}{2} \\
 &\quad \frac{dt}{d\theta} = \frac{1}{2} \sec^2 \frac{\theta}{2} \\
 &\quad = \frac{1}{2} (1 + \tan^2 \frac{\theta}{2}) \\
 &\quad = \frac{1}{2} (1 + t^2) \\
 &\quad 2 dt = (1 + t^2) d\theta \\
 &\quad d\theta = \frac{2 dt}{1 + t^2}
 \end{aligned}$$

$$\text{When } \theta = \frac{\pi}{3}, t = \tan \frac{\pi}{6} = \frac{1}{\sqrt{3}}$$

$$\theta = 0, t = \tan 0 = 0$$

$$\begin{aligned}
 \text{Now } I &= \int_{\frac{1}{\sqrt{3}}}^{\frac{\pi}{2}} \frac{1}{3 - \left(\frac{1-t^2}{1+t^2} \right)} \cdot \frac{2 dt}{1+t^2} \\
 &= \int_{\frac{1}{\sqrt{3}}}^{\frac{\pi}{2}} \frac{2 dt}{(1+t^2)[3(1+t^2) - (1-t^2)]} \\
 &= 2 \int_{\frac{1}{\sqrt{3}}}^{\frac{\pi}{2}} \frac{dt}{3+3t^2-1+t^2} = 2 \int_{\frac{1}{\sqrt{3}}}^{\frac{\pi}{2}} \frac{dt}{4t^2+2} \\
 &= \frac{1}{2} \int_{\frac{1}{\sqrt{3}}}^{\frac{\pi}{2}} \frac{dt}{2t^2+1} = \int_{\frac{1}{\sqrt{3}}}^{\frac{\pi}{2}} \frac{dt}{2t^2+1} \\
 &= \frac{1}{2} \int_{\frac{1}{\sqrt{3}}}^{\frac{\pi}{2}} \frac{dt}{t^2 + \frac{1}{2}} \\
 &= \frac{1}{2} \left[\sqrt{2} \tan^{-1} \sqrt{2} t \right]_{\frac{1}{\sqrt{3}}}^{\frac{\pi}{2}} \\
 &= \frac{\sqrt{2}}{2} \left(\tan^{-1} \sqrt{2} \left(\frac{\sqrt{3}}{2} \right) - \tan^{-1} \sqrt{2}(0) \right) \\
 &\approx 0.484
 \end{aligned}$$



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Further Integration



- (i) Show that $\frac{d}{dx} [(1+x) \ln(1+x) - x] = \ln(1+x)$. 2

Consider the reduction formula $I_n = \int_0^1 x^n \ln(1+x) dx$ for $n = 0, 1, 2, \dots$.

- (ii) Show that $(n+1)I_n = 2 \ln 2 - \frac{1}{n+1} - nI_{n-1}$, $n = 1, 2, \dots$. 3

- (iii) Hence evaluate $4I_3$. 2



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Further Integration



- (i) Find constants A and B such that
 $A(3 \sin x + 2 \cos x) + B(3 \cos x - 2 \sin x) = 8 \sin x + 14 \cos x$ 2

- (ii) Hence, find the exact value of $\int_0^{\frac{\pi}{2}} \frac{8 \sin x + 14 \cos x}{3 \sin x + 2 \cos x} dx$ 3



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Further Integration



- (i) Use the substitution $u = \frac{1}{x}$ to show that $\int_{\frac{1}{a}}^a \frac{\ln x}{1+x^2} dx = 0$ 2

for any constant $a > 0$.

- (ii) Hence, find $\int_{\frac{1}{\sqrt{3}}}^{\sqrt{3}} \frac{\tan^{-1} x}{x} dx$ in simplest exact form. 3



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Further Integration



Let $I_n = \int_0^1 \frac{x^n}{x^2+1} dx$ for $n \geq 2$

- (i) Show that $I_n + I_{n-2} = \frac{1}{n-1}$ 2

- (ii) Hence, or otherwise, show that $\int_0^1 \frac{x^2(x-1)^2}{x^2+1} dx = \ln 2 - \frac{2}{3}$ 3



Further Integration Solution

$$(i) A(3\sin x + 2\cos x) + B(3\cos x - 2\sin x) = 8\sin x + 14\cos x$$

$$3A\sin x + 2A\cos x + 3B\cos x - 2B\sin x = 8\sin x + 14\cos x$$

$$(3A - 2B)\sin x + (2A + 3B)\cos x = 8\sin x + 14\cos x$$

Equating coefficients,

$$3A - 2B = 8 \quad \text{--- (1)}$$

$$2A + 3B = 14 \quad \text{--- (2)}$$

$$(1) \times 2: 6A - 4B = 16 \quad \text{--- (3)}$$

$$(2) \times 3: 6A + 9B = 42 \quad \text{--- (4)}$$

$$(4) - (3): 13B = 26$$

$$B = 2 \quad \text{--- (5)}$$

$$\text{Sub (5) in (1): } 3A - 2(2) = 8$$

$$3A = 12$$

$$A = 4$$

$$\therefore A = 4, B = 2$$

$$(ii) \int_0^{\pi/2} \frac{8\sin x + 14\cos x}{3\sin x + 2\cos x} dx$$

$$= \int_0^{\pi/2} \frac{4(3\sin x + 2\cos x) + 2(3\cos x - 2\sin x)}{3\sin x + 2\cos x} dx$$

$$= \int_0^{\pi/2} 4 + \frac{2(3\cos x - 2\sin x)}{3\sin x + 2\cos x} dx$$

$$= \left[4x + 2\ln|3\sin x + 2\cos x| \right]_0^{\pi/2}$$

$$= \frac{4\pi}{2} + 2\ln|3(1) + 2(0)| - (0 + 2\ln|3(0) + 2(1)|)$$

$$= 2\pi + 2\ln(3/2)$$



Further Integration Solution

$$i. \frac{d}{dx} [(1+x)\ln(1+x) - x]$$

$$\text{Let } u = 1+x \quad v = \ln(1+x)$$

$$u' = 1 \quad v' = \frac{1}{1+x}$$

$$= vu' + uv' - 1$$

$$= 1 \times \ln(1+x) + (1+x)\left(\frac{1}{1+x}\right) - 1$$

$$= \ln(1+x) + \frac{1+x}{1+x} - 1$$

$$= \ln(1+x) + 1 - 1$$

$$= \ln(1+x)$$

$$ii. I_n = \int_0^1 x^n \ln(1+x) dx$$

$$\text{Let } u = x^n \quad v = (1+x)\ln(1+x) - x$$

$$u' = nx^{n-1} \quad v' = \ln(1+x)$$

$$= \left[x^n \{(1+x)\ln(1+x) - x\} \right]_0^1 - \int_0^1 [nx^{n-1} \{(1+x)\ln(1+x) - x\}] dx$$

$$= 1 \{2\ln 2 - 1\} - n \int_0^1 \{x^{n-1} \ln(1+x) + x^n \ln(1+x) - x^n\} dx$$

$$= 2\ln 2 - 1 - n \int_0^1 x^{n-1} \ln(1+x) dx - n \int_0^1 x^n \ln(1+x) dx + n \int_0^1 x^n dx$$

$$= 2\ln 2 - 1 - nI_{n-1} - nI_n + n \left[\frac{x^{n+1}}{n+1} \right]_0^1$$

$$= 2\ln 2 - 1 - nI_{n-1} - nI_n + \frac{n}{n+1}$$

$$\therefore nI_n + I_n = 2\ln 2 - nI_{n-1} - 1 + \frac{n}{n+1}$$

$$= 2\ln 2 - nI_{n-1} - \frac{n+1}{n+1} + \frac{n}{n+1}$$

$$= 2\ln 2 - nI_{n-1} - \frac{1+n-n}{n+1}$$

$$= 2\ln 2 - \frac{1}{n+1} - nI_{n-1}$$

$$iii. 4I_3 = 2\ln 2 - \frac{1}{4} - 3I_2 \quad (\text{from part ii})$$

$$= 2\ln 2 - \frac{1}{4} - [2\ln 2 - \frac{1}{3} - 2I_1]$$

$$= 2\ln 2 - \frac{1}{4} - 2\ln 2 + \frac{1}{3} + [2\ln 2 - \frac{1}{2} - I_0]$$

$$= 2\ln 2 - \frac{1}{4} - 2\ln 2 + \frac{1}{3} + 2\ln 2 - \frac{1}{2} - \int_0^1 \ln(1+x) dx$$

$$= 2\ln 2 - \frac{1}{4} + \frac{1}{3} - \frac{1}{2} - \left[(1+x)\ln(1+x) - x \right]_0^1$$

$$= 2\ln 2 - \frac{1}{4} + \frac{1}{3} - \frac{1}{2} - [2\ln 2 - 1 - (1(1) - 0)] = 1 - \frac{1}{4} + \frac{1}{3} - \frac{1}{2} = \frac{7}{12}$$



Further Integration Solution

$$(i) LHS = I_n + I_{n-2}$$

$$= \int_0^1 \frac{x^n}{x^2+1} dx + \int_0^1 \frac{x^{n-2}}{x^2+1} dx$$

$$= \int_0^1 \frac{x^{n-2}(x^2+1)}{x^2+1} dx$$

$$= \left[\frac{x^{n-1}}{n-1} \right]_0^1 = \frac{1}{n-1} - 0 = \frac{1}{n-1} = RHS$$

$$(ii) \int_0^1 \frac{x^2(x-1)^2}{x^2+1} dx = \int_0^1 \frac{x^2(x^2-2x+1)}{x^2+1} dx$$

$$= \int_0^1 \frac{x^4}{x^2+1} - \frac{2x^3}{x^2+1} + \frac{x^2}{x^2+1} dx$$

$$= I_4 - 2I_3 + I_2$$

$$= \frac{1}{5} - 2\left(\frac{1}{3} - I_1\right) \quad \text{from (i)}$$

$$= \frac{1}{5} - 1 + \int_0^1 \frac{2x}{x^2+1} dx$$

$$= -\frac{2}{5} + \left[\ln(x^2+1) \right]_0^1$$

$$= -\frac{2}{5} + \ln 2 - \ln 1 = \ln 2 - \frac{2}{5}$$



Further Integration Solution

$$\int_{\frac{1}{2}}^1 \frac{\ln x}{1+x^2} dx$$

$$\text{Let } u = \frac{1}{x} \rightarrow x = \frac{1}{u}$$

$$du = -\frac{1}{u^2} dx$$

$$dx = -u^2 du$$

$$dx = -\frac{1}{u^2} du$$

$$\begin{cases} x = 1/2, u = 2 \\ x = 1, u = 1 \end{cases}$$

$$= \int_2^1 \frac{-\ln u}{u^2+1} \cdot \frac{1}{u^2} du$$

$$= \int_1^2 \frac{-\ln u}{u^2+1} du$$

$$= \int_{\frac{1}{2}}^1 \frac{-\ln x}{1+x^2} dx \quad \text{Dummy variable}$$

$$2 \int_{\frac{1}{2}}^1 \frac{\ln x}{1+x^2} dx = 0$$

$$\therefore \int_{\frac{1}{2}}^1 \frac{\ln x}{1+x^2} dx = 0$$

$$(ii) \int_{\frac{1}{\sqrt{3}}}^{\sqrt{3}} \frac{\tan^{-1} x}{x} dx$$

$$u = \tan^{-1} x \quad v' = \frac{1}{x}$$

$$u' = \frac{1}{1+x^2} \quad v = \ln x$$

$$= \left[\ln x \cdot \tan^{-1} x \right]_{\frac{1}{\sqrt{3}}}^{\sqrt{3}} - \int_{\frac{1}{\sqrt{3}}}^{\sqrt{3}} \frac{\ln x}{1+x^2} dx$$

$$= \ln \sqrt{3} \cdot \frac{\pi}{3} - \ln\left(\frac{1}{\sqrt{3}}\right) \cdot \frac{\pi}{6} - 0 \quad \text{from (i)}$$

$$= \frac{\pi \ln \sqrt{3}}{3} + \frac{\pi \ln \sqrt{3}}{6} = \frac{\pi \ln \sqrt{3}}{2} = \frac{\pi \ln 3}{4}$$



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Further Integration



Consider $I_n = \int x^m \ln^n x \, dx$

(i) Show that $I_n = \frac{x^{m+1}}{m+1} \ln^n x - \frac{n}{m+1} I_{n-1}$ 2

(ii) Hence, evaluate $I_n = \int_1^2 x^3 \ln^4 x \, dx$ 3



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Further Integration



For $n = 0, 1, 2, \dots$ define

$$I_n = \int_0^{\pi/4} \tan^n \theta \, d\theta.$$

(i) Show that $I_1 = \frac{1}{2} \ln 2$. 1

(ii) Show that for $n \geq 2$, 3

$$I_n + I_{n-2} = \frac{1}{n-1}.$$

(iii) For $n \geq 2$, explain why $I_n < I_{n-2}$, and deduce that 3

$$\frac{1}{2(n+1)} < I_n < \frac{1}{2(n-1)}.$$

(iv) By using the recurrence relation in part (ii), find I_5 and deduce that 2

$$\frac{2}{3} < \ln 2 < \frac{3}{4}.$$



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Further Integration



Let $I_n = \int_0^1 \frac{x^n}{x+1} \, dx$ for $n \geq 0$.

(i) Use algebraic manipulation to show that $I_n = \frac{1}{n} - I_{n-1}$ for $n \geq 1$. 2

(ii) Hence find I_2 . 2



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Further Integration



(i) The function $f(x)$ is continuous and differentiable for all real x . 2

Use the substitution $u = a + b - x$ to show that

$$\int_a^b \frac{f(x)}{f(a+b-x) + f(x)} \, dx = \frac{b-a}{2}.$$

(ii) Hence use the substitution $u = x^2$ to evaluate $\int_1^3 \frac{x^2}{\sqrt{10-x^2} + x} \, dx$. 2



Further Integration Solution

$$\begin{aligned} \text{(i)} \quad I_1 &= \int_0^{\frac{\pi}{4}} \tan x \, dx = \left[-\ln|\cos x| \right]_0^{\frac{\pi}{4}} \\ &= \ln|1| - \ln\left|\frac{1}{\sqrt{2}}\right| = \ln|\sqrt{2}| = \frac{1}{2} \ln 2 \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad &\text{Either } I_{n+1} + I_{n-2} \\ &= \int_0^{\frac{\pi}{4}} \tan^n x \, dx + \int_0^{\frac{\pi}{4}} \tan^{n-2} x \, dx \\ &= \int_0^{\frac{\pi}{4}} \tan^{n-2} x (\tan^2 x + 1) \, dx \\ &= \int_0^{\frac{\pi}{4}} \tan^{n-2} x \sec^2 x \, dx = \left[\frac{\tan^{n-1} x}{n-1} \right]_0^{\frac{\pi}{4}} = \frac{1}{n-1} \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad &\text{Since } x \in [0, \frac{\pi}{4}] \\ &\text{then } \tan x \in [0, 1] \\ &\therefore \int_0^{\frac{\pi}{4}} \tan^n x \, dx < \int_0^{\frac{\pi}{4}} \tan^{n-1} x \, dx \end{aligned}$$

Since if a fraction raised to a power is smaller than fraction raised to a smaller power

$$\left(\frac{1}{2}\right)^3 < \left(\frac{1}{2}\right)$$

$$\begin{aligned} \text{(iv)} \quad &I_n < I_{n-2} \\ &2I_n < I_n + I_{n-2} \\ &I_n + I_{n-2} = \frac{1}{n-1} \quad \text{part (ii)} \\ &I_n < \frac{1}{2(n-1)} \\ &I_{n+2} < I_n \quad (\text{map } n \text{ to } n+2) \\ &I_{n+2} + I_n < 2I_n \\ &\frac{1}{n+2-1} < 2I_n \\ &\therefore \frac{1}{2(n+1)} < I_n < \frac{1}{2(n-1)} \end{aligned}$$



Further Integration Solution

$$\begin{aligned} \text{(i)} \quad &u = \ln^n x \quad v' = x^m \\ &u' = \frac{n \ln^{n-1} x}{x} \quad v = \frac{x^{m+1}}{m+1} \\ &\int u v' \, dx = uv - \int u' v \, dx \\ &\int x^m \ln^n x \, dx = \frac{x^{m+1}}{m+1} \ln^n x - \int \frac{x^{m+1}}{m+1} \frac{n \ln^{n-1} x}{x} \, dx \\ &= \frac{x^{m+1}}{m+1} \ln^n x - \frac{n}{m+1} \int x^m \ln^{n-1} x \, dx \\ &= \frac{x^{m+1}}{m+1} \ln^n x - \frac{n}{m+1} I_{n-1} \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad &\int_1^2 x^3 \ln^4 x \, dx \\ &= \left[\frac{x^4}{4} \ln^4 x \right]_1^2 - I_3 \\ &= \left[\frac{2^4}{4} \ln^4 2 - 0 \right] - \left\{ \left[\frac{x^4}{4} \ln^3 x \right]_1^2 - \frac{3}{4} I_2 \right\} \\ &= 4 \ln^4 2 - \left[\frac{2^4}{4} \ln^3 2 - 0 \right] + \frac{3}{4} \left\{ \left[\frac{x^4}{4} \ln^2 x \right]_1^2 - \frac{2}{4} I_1 \right\} \\ &= 4 \ln^4 2 - 4 \ln^3 2 + \frac{3}{4} \left[\frac{2^4}{4} \ln^2 2 - 0 \right] - \frac{3}{4} \times \frac{2}{4} \left\{ \left[\frac{x^4}{4} \ln x \right]_1^2 - \frac{1}{4} I_0 \right\} \\ &= 4 \ln^4 2 - 4 \ln^3 2 + 3 \ln^2 2 - \frac{3}{4} \times \frac{2}{4} \left[\frac{2^4}{4} \ln 2 - 0 \right] + \frac{3}{4} \times \frac{2}{4} \times \frac{1}{4} \int_1^2 x^3 \, dx \\ &= 4 \ln^4 2 - 4 \ln^3 2 + 3 \ln^2 2 - \frac{3}{2} \ln 2 + \frac{3}{32} \left[\frac{x^4}{4} \right]_1^2 \\ &= 4 \ln^4 2 - 4 \ln^3 2 + 3 \ln^2 2 - \frac{3}{2} \ln 2 + \frac{3}{32} \left[4 - \frac{1}{4} \right] \\ &= 4 \ln^4 2 - 4 \ln^3 2 + 3 \ln^2 2 - \frac{3}{2} \ln 2 + \frac{45}{128} \end{aligned}$$



Further Integration Solution

$$\begin{aligned} \text{(i)} \quad I &= \int_a^b \frac{f(x)}{f(a+b-x) + f(x)} \, dx \quad \text{let } u = a+b-x \\ &\quad \quad \quad du = -dx \\ &\quad \quad \quad x=a, \quad u=b \\ &\quad \quad \quad x=b, \quad u=a \\ &= \int_b^a \frac{f(a+b-u)}{f(u) + f(a+b-u)} \, -du \\ &= \int_a^b \frac{f(a+b-u)}{f(u) + f(a+b-u)} \, du \quad \checkmark \\ &= \int_a^b \frac{f(a+b-x)}{f(a+b-x) + f(x)} \, dx \quad (\text{dummy variable}) \\ \therefore 2I &= \int_a^b \frac{f(x)}{f(a+b-x) + f(x)} \, dx + \int_a^b \frac{f(a+b-x)}{f(a+b-x) + f(x)} \, dx \\ &= \int_a^b \frac{f(x) + f(a+b-x)}{f(a+b-x) + f(x)} \, dx \\ &= \int_a^b 1 \, dx \\ &= b-a \\ I &= \frac{b-a}{2} \quad \checkmark \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad &\int_1^3 \frac{x^2 \, dx}{\sqrt{10-x^2} + x} \quad \text{let } u = x^2 \\ &\quad \quad \quad du = 2x \, dx \\ &\quad \quad \quad x=1, \quad u=1 \\ &\quad \quad \quad x=3, \quad u=9 \\ &= \frac{1}{2} \int_1^9 \frac{x \cdot 2x \, dx}{\sqrt{10-x^2} + x} \\ &= \frac{1}{2} \int_1^9 \frac{\sqrt{u} \, du}{\sqrt{10-u} + \sqrt{u}} \quad \checkmark \\ &= \frac{1}{2} \times \frac{9-1}{2} \quad \text{from (i)} = 2 \end{aligned}$$



Further Integration Solution

$$I_n = \int_0^1 \frac{x^n}{x+1} \, dx, \quad n \geq 0$$

$$\begin{aligned} \text{(i)} \quad I_n &= \int_0^1 \frac{(x+1)x^{n-1} - x^{n-1}}{x+1} \, dx \\ &= \int_0^1 \left(x^{n-1} - \frac{x^{n-1}}{x+1} \right) \, dx \\ &= \left[\frac{x^n}{n} \right]_0^1 - I_{n-1} = \frac{1}{n} - I_{n-1} \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad I_0 &= \int_0^1 \frac{dx}{x+1} \\ &= [\ln(x+1)]_0^1 \\ &= \ln 2 - \ln 1 \\ &= \ln 2 \end{aligned}$$

$$\therefore I_1 = \frac{1}{1} - \ln 2$$

$$= 1 - \ln 2$$

$$I_2 = \frac{1}{2} - (1 - \ln 2)$$

$$= \ln 2 - \frac{1}{2}$$



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Further Integration



Consider the two integrals $I = \int_{-1}^1 \frac{1}{(1+x^2)^2} dx$ and $J = -\int_{-1}^1 \frac{t^2}{(1+t^2)^2} dt$.

- (i) Explain why the use of the substitution $x = \frac{1}{t}$ does not demonstrate that $I = J$. 1
- (ii) Evaluate I . 2
- (iii) Evaluate J . 2



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Further Integration



Evaluate the following definite integrals:

(i) $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} e^x \sin x \, dx$. 3

(ii) $\int_{-1}^1 \frac{\sin \frac{2\pi}{5}}{x^2 + 2x \cos \frac{2\pi}{5} + 1} dx$. 3

(Hint: Express the denominator in the form $(x+a)^2 + b^2$)



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Further Integration



Find the integral $\int \sec^3 x \, dx$ 3



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Further Integration



Evaluate $\int_{\frac{1}{3}}^1 \sqrt{x} \tan^{-1} \sqrt{x} \, dx$. 3



Further Integration Solution

$$\begin{aligned}
 \text{(i) Let } I &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} e^x \sin x \, dx \\
 I &= \left[e^x \sin x \right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} - \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} e^x \cos x \, dx \\
 &= e^{\frac{\pi}{2}} \sin \frac{\pi}{2} - e^{-\frac{\pi}{2}} \sin \left(-\frac{\pi}{2}\right) - \left[e^x \cos x \right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} + \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} e^x (-\sin x) \, dx \\
 &= e^{\frac{\pi}{2}} + e^{-\frac{\pi}{2}} - e^{\frac{\pi}{2}} \cos \frac{\pi}{2} - e^{-\frac{\pi}{2}} \cos \left(-\frac{\pi}{2}\right) - I \\
 2I &= e^{\frac{\pi}{2}} + e^{-\frac{\pi}{2}} \\
 I &= \frac{e^{\frac{\pi}{2}} + e^{-\frac{\pi}{2}}}{2}
 \end{aligned}$$

NOTE:

A common mistake was to not apply the fact that $\sin(-\frac{\pi}{2}) = -1$ and pupils ended up with $\frac{e^{\frac{\pi}{2}} - e^{-\frac{\pi}{2}}}{2}$ as an answer.

Some pupils did not even apply the bounds to the primitive of certain parts they found and others mistakenly put $\frac{\pi}{2}$ instead of $-\frac{\pi}{2}$ for the lower bound in the solution.

$$\begin{aligned}
 \text{(ii) } \int_{-1}^1 \frac{\sin \frac{2x}{5}}{x^2 + 2x \cos \frac{2\pi}{5} + 1} \, dx \\
 \text{(Hint: Express the denominator in the form } (x+a)^2 + b^2) \\
 \int_{-1}^1 \frac{\sin \frac{2x}{5}}{x^2 + 2x \cos \frac{2\pi}{5} + 1} \, dx = \int_{-1}^1 \frac{\sin \frac{2x}{5}}{(x + \cos \frac{2\pi}{5})^2 + \sin^2 \frac{2\pi}{5}} \, dx \\
 = \left[\tan^{-1} \left(\frac{x + \cos \frac{2\pi}{5}}{\sin \frac{2\pi}{5}} \right) \right]_{-1}^1 = \tan^{-1} \left(\frac{1 + \cos \frac{2\pi}{5}}{\sin \frac{2\pi}{5}} \right) - \tan^{-1} \left(\frac{-1 + \cos \frac{2\pi}{5}}{\sin \frac{2\pi}{5}} \right) \\
 = \tan^{-1} \left(\frac{1 + 2 \cos^2 \frac{\pi}{5} - 1}{2 \sin \frac{\pi}{5} \cos \frac{\pi}{5}} \right) - \tan^{-1} \left(\frac{-1 + 1 - 2 \sin^2 \frac{\pi}{5}}{2 \sin \frac{\pi}{5} \cos \frac{\pi}{5}} \right) \\
 = \tan^{-1} \left(\cot \frac{\pi}{5} \right) - \tan^{-1} \left(\tan \left(-\frac{\pi}{5}\right) \right) \\
 = \tan^{-1} \left(\tan \left(\frac{\pi}{2} - \frac{\pi}{5}\right) \right) - \tan^{-1} \left(\tan \left(-\frac{\pi}{5}\right) \right) = \frac{\pi}{2} - \frac{\pi}{5} - \left(-\frac{\pi}{5}\right) = \frac{\pi}{2}
 \end{aligned}$$

NOTE:

Many pupils put as their answer, the primitive with the bounds applied and went no further. Thus they could not receive full marks.



Further Integration Solution

- (i) When x is zero, t is infinite which means this substitution should be done with 2 integrals with an infinite boundary on each. Another valid response is that the substitution $x = \frac{1}{t}$ must be continuous and one to one over the interval of x .

NOTE: Pupils attempted to use the substitution to show these are not correct, but they should've ended up with $I = J$. The question states it's not correct and wants pupils to explain why. Many pupils wasted time on this and failed to address the question in any way. Some pupils gave a vague or generic response and if this were the case, could not receive full marks without giving more details to explain their position.

$$\text{(ii) The integrand is an even function, hence } I = 2 \int_0^1 \frac{1}{(1+x^2)^2} \, dx$$

$$\text{Let } x = \tan \theta, \frac{dx}{d\theta} = \sec^2 \theta, \text{ when } x = 1, \theta = \frac{\pi}{4} \text{ and when } x = 0, \theta = 0.$$

$$\begin{aligned}
 I &= 2 \int_0^{\frac{\pi}{4}} \frac{1}{(1 + \tan^2 \theta)^2} \sec^2 \theta \, d\theta = 2 \int_0^{\frac{\pi}{4}} \frac{1}{\sec^2 \theta} \, d\theta = 2 \int_0^{\frac{\pi}{4}} \cos^2 \theta \, d\theta \\
 &= 2 \int_0^{\frac{\pi}{4}} \frac{1 + \cos 2\theta}{2} \, d\theta = \left[\theta + \frac{\sin 2\theta}{2} \right]_0^{\frac{\pi}{4}} = \frac{\pi}{4} + \frac{\sin \frac{\pi}{2}}{2} - \left(0 + \frac{\sin 0}{2} \right) = \frac{\pi}{4} + \frac{1}{2}
 \end{aligned}$$

$$\text{(iii) The integrand is an even function, hence } J = -2 \int_0^1 \frac{t^2}{(1+t^2)^2} \, dt$$

$$\text{Let } x = \tan \theta, \frac{dx}{d\theta} = \sec^2 \theta, \text{ when } x = 1, \theta = \frac{\pi}{4} \text{ and when } x = 0, \theta = 0.$$

$$\begin{aligned}
 J &= -2 \int_0^{\frac{\pi}{4}} \frac{\tan^2 \theta}{(1 + \tan^2 \theta)^2} \sec^2 \theta \, d\theta = -2 \int_0^{\frac{\pi}{4}} \frac{\tan^2 \theta}{\sec^2 \theta} \, d\theta = -2 \int_0^{\frac{\pi}{4}} \sin^2 \theta \, d\theta \\
 &= -2 \int_0^{\frac{\pi}{4}} \frac{1 - \cos 2\theta}{2} \, d\theta = \left[\frac{\sin 2\theta}{2} - \theta \right]_0^{\frac{\pi}{4}} = \frac{\sin \frac{\pi}{2}}{2} - \frac{\pi}{4} - \left(\frac{\sin 0}{2} - 0 \right) = \frac{1}{2} - \frac{\pi}{4}
 \end{aligned}$$

NOTE:

Some pupils tried to do this without a trigonometric substitution, however none were successful in doing so. A few pupils tried $x = \sin \theta$, but this gave $1 + \sin^2 \theta$ in the denominator which doesn't end up being particularly useful. This should have been an indicator for them to try a different substitution, but all who did this pushed through with their substitution and failed to get full marks.



Further Integration Solution

$$\begin{aligned}
 \int_{\frac{1}{3}}^1 \sqrt{x} \tan^{-1} \sqrt{x} \, dx &= [uv]_{\frac{1}{3}}^1 - \int_{\frac{1}{3}}^1 u'v \, dx \\
 \text{where } u &= \tan^{-1} \left(x^{\frac{1}{2}} \right) \text{ and } v' = x^{\frac{1}{2}} \\
 u' &= \frac{\frac{1}{2} x^{-\frac{1}{2}}}{1 + (\sqrt{x})^2} & v &= \frac{2x^{\frac{3}{2}}}{3} \\
 u' &= \frac{2\sqrt{x}}{1+x} & & \\
 &= \frac{2}{2\sqrt{x}(1+x)} \\
 &= \frac{2}{3} \left[x\sqrt{x} \tan^{-1} \sqrt{x} \right]_{\frac{1}{3}}^1 - \frac{2}{3} \int_{\frac{1}{3}}^1 \frac{1}{2\sqrt{x}(1+x)} (x\sqrt{x}) \, dx \\
 &= \frac{2}{3} \left[(1)\sqrt{(1)} \tan^{-1} \sqrt{(1)} - \left(\frac{1}{3}\right)\sqrt{\left(\frac{1}{3}\right)} \tan^{-1} \sqrt{\left(\frac{1}{3}\right)} \right] - \frac{1}{3} \int_{\frac{1}{3}}^1 \frac{x}{1+x} \, dx \\
 &= \frac{2}{3} \left[\tan^{-1} 1 - \frac{1}{3} \left(\frac{1}{\sqrt{3}} \right) \tan^{-1} \frac{1}{\sqrt{3}} \right] - \frac{1}{3} \int_{\frac{1}{3}}^1 \frac{(x+1)-1}{1+x} \, dx \\
 &= \frac{2}{3} \left[\frac{\pi}{4} - \frac{1}{3\sqrt{3}} \left(\frac{\pi}{6} \right) \right] - \frac{1}{3} \int_{\frac{1}{3}}^1 1 - \frac{1}{1+x} \, dx \\
 &= \frac{\pi}{6} - \frac{\pi}{27\sqrt{3}} - \frac{1}{3} [x - \ln|1+x|]_{\frac{1}{3}}^1 \\
 &= \frac{\pi}{6} - \frac{\pi}{27\sqrt{3}} - \frac{1}{3} \left[(1) - \ln|1| + \left(\frac{1}{3}\right) - \ln\left|1 + \left(\frac{1}{3}\right)\right| \right] \\
 &= \frac{\pi}{6} - \frac{\pi}{27\sqrt{3}} - \frac{1}{3} \left[1 - \ln 2 - \left(\frac{1}{3}\right) + \ln \frac{4}{3} \right] \\
 &= \frac{\pi}{6} - \frac{\pi}{27\sqrt{3}} - \frac{1}{3} \left[\frac{2}{3} + \ln \left(\frac{4}{3} \right) \right] \\
 &= \frac{\pi}{6} - \frac{\pi}{27\sqrt{3}} - \frac{1}{3} \left[\frac{2}{3} + \ln \left(\frac{8}{3} \right) \right] \\
 &= \frac{\pi}{6} - \frac{\pi}{27\sqrt{3}} - \frac{2}{9} - \frac{1}{3} \ln \left(\frac{8}{3} \right)
 \end{aligned}$$



Further Integration Solution

$$\begin{aligned}
 \text{Let } I &= \int \sec^3 x \, dx = \int \sec^2 x \sec x \, dx \\
 &= uv - \int u'v \, dx \text{ where } u = \sec x \text{ and } v' = \sec^2 x \\
 &= \sec x \tan x - \int \sec x \tan^2 x \, dx \\
 &= \sec x \tan x - \int \sec x (\sec^2 x - 1) \, dx \\
 &= \sec x \tan x - \int (\sec^3 x - \sec x) \, dx \\
 &= \sec x \tan x - \int \sec^3 x \, dx + \int \sec x \, dx \\
 I &= \sec x \tan x - I + \int \sec x \times \frac{\sec x + \tan x}{\sec x + \tan x} \, dx \\
 2I &= \sec x \tan x + \int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x} \, dx \\
 2I &= \sec x \tan x + \ln|\sec x + \tan x| \\
 \therefore I &= \frac{1}{2} \sec x \tan x + \frac{1}{2} \ln|\sec x + \tan x| + C
 \end{aligned}$$



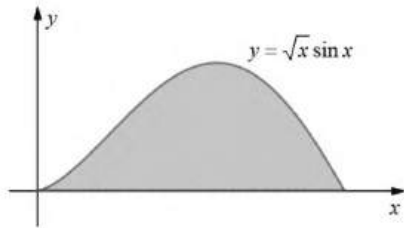
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Further Integration



- (i) Find $\int \sin^2 x \, dx$ 2
- (ii) Show that $\int x \sin^2 x \, dx = \frac{1}{8}(2x^2 - 2x \sin 2x - \cos 2x) + C$ 3
- (iii) The diagram shows a finite region bound by the curve $y = \sqrt{x} \sin x$ and the x axis.



Find the volume of revolution when the shaded region is rotated about the x axis. 3



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Further Integration



The integral I_n is defined by $I_n = \int_0^1 e^x (2-x)^n \, dx$ for integers $n \geq 0$.

- (i) Show that $I_n = e - 2^n + nI_{n-1}$ 2
- (ii) Hence or otherwise, evaluate $\int_0^1 e^x (2-x)^4 \, dx$ 3



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Further Integration



- (i) Show that: 2
- $$\int_0^a f(x) \, dx = \int_0^a f(a-x) \, dx$$
- (ii) Hence, or otherwise, evaluate: 3

$$\int_0^{\frac{\pi}{2}} \frac{\sin x}{\cos x + \sin x} \, dx$$



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Further Integration



Find $\int \frac{\ln x - 1}{[x + \ln x]^2} \, dx$. 3



Further Integration Solution

$$\begin{aligned}
 \text{(i)} \quad & \text{Let } u = (2-x)^n \quad v = e^x \\
 & u' = -n(2-x)^{n-1} \quad v' = e^x \\
 I_n &= [uv]_0^1 - \int_0^1 vu' dx \\
 &= \left[e^x (2-x)^n \right]_0^1 - \int_0^1 -e^x n(2-x)^{n-1} dx \\
 &= e^1 (2-1)^n - e^0 (2-0)^n + n \int_0^1 e^x (2-x)^{n-1} dx \\
 &= e - 2^n + nI_{n-1}
 \end{aligned}$$

(ii)

Required to find I_4

$$\begin{aligned}
 I_0 &= \int_0^1 e^x dx = [e^x]_0^1 = e - 1 \\
 I_1 &= e - 2^1 + I_0 = e - 2 + e - 1 = 2e - 3 \\
 I_2 &= e - 2^2 + 2I_1 = e - 4 + 2(2e - 3) = 5e - 10 \\
 I_3 &= e - 2^3 + 3I_1 = e - 8 + 3(5e - 10) = 16e - 38 \\
 I_4 &= e - 2^4 + 4I_1 = e - 16 + 4(16e - 38) = 65e - 168
 \end{aligned}$$



Further Integration Solution

$$\begin{aligned}
 \text{(i)} \quad & \cos 2x = 1 - 2\sin^2 x \quad \int \sin^2 x dx = \frac{1}{2} \int (1 - \cos 2x) dx \\
 & \sin^2 x = \frac{1}{2}(1 - \cos 2x) = \frac{1}{2} \left(x - \frac{1}{2} \sin 2x \right) + C
 \end{aligned}$$

$$\text{(ii)} \quad \text{Let } u = x \quad v = \frac{1}{2} \left(x - \frac{1}{2} \sin 2x \right) \text{ from (i)}$$

$$u' = 1 \quad v' = \sin^2 x$$

$$\begin{aligned}
 \int x \sin^2 x dx &= uv - \int vu' dx \\
 &= \frac{1}{2} x \left(x - \frac{1}{2} \sin 2x \right) - \frac{1}{2} \int \left(x - \frac{1}{2} \sin 2x \right) dx \\
 &= \frac{x^2}{2} - \frac{x \sin 2x}{4} - \frac{1}{2} \left(\frac{x^2}{2} + \frac{1}{4} \cos 2x \right) + C \\
 &= \frac{x^2}{2} - \frac{x \sin 2x}{4} - \frac{x^2}{4} - \frac{\cos 2x}{8} + C \\
 &= \frac{1}{8} (4x^2 - 2x \sin 2x - 2x^2 - \cos 2x) + C \\
 &= \frac{1}{8} (2x^2 - 2x \sin 2x - \cos 2x) + C
 \end{aligned}$$

$$\text{(iii)} \quad x \text{ intercepts: } \sqrt{x} \sin x = 0 \quad x = 0, \pi$$

Volume of revolution:

$$\begin{aligned}
 V &= \pi \int_0^\pi x \sin^2 x dx \\
 &= \frac{\pi}{8} [2x^2 - 2x \sin 2x - \cos 2x]_0^\pi \\
 &= \frac{\pi}{8} [2\pi^2 - 0 - 1 - 0 + 0 + 1] = \frac{\pi^3}{4}
 \end{aligned}$$



Further Integration Solution

$$\begin{aligned}
 I &= \int \frac{\ln x - 1}{[x + \ln x]^2} dx \\
 &= \int \frac{x + \ln x - x - 1}{[x + \ln x]^2} dx \\
 &= \int \frac{x + \ln x}{[x + \ln x]^2} dx - \int \frac{x + 1}{[x + \ln x]^2} dx \\
 &= \int \frac{1}{x + \ln x} dx - \int \frac{x(1 + \frac{1}{x})}{[x + \ln x]^2} dx \\
 &= \int \frac{1}{x + \ln x} dx - J
 \end{aligned}$$

$$\text{Let } J = \int \frac{x(1 + \frac{1}{x})}{[x + \ln x]^2} dx$$

By using integration by parts on J

$$u = x \quad v' = \frac{(1 + \frac{1}{x})}{[x + \ln x]^2}$$

$$u' = 1 \quad v = \frac{-1}{x + \ln x} \quad \text{by Reverse Chain Rule}$$

$$\begin{aligned}
 J &= x \left(\frac{-1}{x + \ln x} \right) - \int \frac{-1}{x + \ln x} dx \\
 &= \frac{-x}{x + \ln x} + \int \frac{1}{x + \ln x} dx
 \end{aligned}$$

$$\begin{aligned}
 &= \int \frac{1}{x + \ln x} dx - \left(\frac{-x}{x + \ln x} + \int \frac{1}{x + \ln x} dx \right) \\
 &= \frac{x}{x + \ln x} + C
 \end{aligned}$$



Further Integration Solution

(i) Let $u = a - x$. Then:

$$\begin{aligned}
 \int_0^a f(x) dx &= - \int_a^0 f(a - u) du \\
 &= \int_0^a f(a - x) dx \quad (u \text{ is a dummy variable})
 \end{aligned}$$

(ii) Let

$$\begin{aligned}
 I &= \int_0^{\frac{\pi}{2}} \frac{\sin x}{\cos x + \sin x} dx \\
 I &= \int_0^{\frac{\pi}{2}} \frac{\sin(\frac{\pi}{2} - x)}{\cos(\frac{\pi}{2} - x) + \sin(\frac{\pi}{2} - x)} dx \\
 I &= \int_0^{\frac{\pi}{2}} \frac{\cos x}{\sin x + \cos x} dx
 \end{aligned}$$

Adding the first and last lines:

$$\begin{aligned}
 2I &= \int_0^{\frac{\pi}{2}} \frac{\sin x + \cos x}{\cos x + \sin x} dx \\
 I &= \frac{1}{2} \int_0^{\frac{\pi}{2}} dx \\
 I &= \frac{\pi}{4}
 \end{aligned}$$



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Further Integration



Let $I_n = \int_0^1 \frac{x^{2n}}{1+x^2} dx$, where n is an integer and $n \geq 0$.

(i) Show that $I_n + I_{n-1} = \frac{1}{2n-1}$. 2

(ii) Hence evaluate $\int_0^1 \frac{x^4}{1+x^2} dx$. 2



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Further Integration



For $n = 0, 1, 2, \dots$, let $I_n = \int_0^{\frac{\pi}{4}} \tan^n \theta d\theta$.

(i) Show that $I_1 = \frac{1}{2} \ln 2$. 1

(ii) Show that for $n \geq 2$, $I_n + I_{n-2} = \frac{1}{n-1}$. 2

(iii) For $n \geq 2$, explain why $I_n < I_{n-2}$, and deduce that $\frac{1}{2(n+1)} < I_n < \frac{1}{2(n-1)}$. 3

(iv) By using the recurrence relation of part (ii), find I_5 and deduce that $\frac{2}{3} < \ln 2 < \frac{3}{4}$. 2



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Further Integration



Find $\int \frac{\sqrt{25x^2 - 4}}{x} dx$. 3



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Further Integration



(i) Prove that $\sqrt{x}(1-\sqrt{x})^{n-1} = (1-\sqrt{x})^{n-1} - (1-\sqrt{x})^n$. 1

(ii) Let $I_n = \int_0^1 (1-\sqrt{x})^n dx$, where $n = 1, 2, 3, \dots$.
show that $I_n = \frac{n}{n+2} I_{n-1}$. 3

(iii) Hence evaluate I_{2022} . 2



Further Integration Solution

$$(i) I_n = \int_0^{\frac{\pi}{4}} \tan^n \theta d\theta$$

$$\begin{aligned} I_1 &= \int_0^{\frac{\pi}{4}} \tan \theta d\theta \\ &= [-\ln(\cos \theta)]_0^{\frac{\pi}{4}} \\ &= \ln[\cos(0)] - \ln\left[\cos\left(\frac{\pi}{4}\right)\right] \\ &= \ln 1 - \ln\left(\frac{1}{\sqrt{2}}\right) \\ &= 0 - \ln\left(2^{-\frac{1}{2}}\right) \\ &= \frac{1}{2} \ln 2 \quad \text{as required.} \end{aligned}$$

$$(ii) \text{ Show that for } n \geq 2, I_n + I_{n-2} = \frac{1}{n-1}.$$

$$\begin{aligned} I_n &= \int_0^{\frac{\pi}{4}} \tan^{n-2} \theta (\sec^2 \theta - 1) d\theta, \text{ where } n \geq 2 \quad (iv) \\ &= \int_0^{\frac{\pi}{4}} \tan^{n-2} \theta \sec^2 \theta d\theta - \int_0^{\frac{\pi}{4}} \tan^{n-2} \theta d\theta \\ &= \int_0^{\frac{\pi}{4}} \tan^{n-2} \theta \sec^2 \theta d\theta - I_{n-2} \\ \therefore I_n + I_{n-2} &= \int_0^{\frac{\pi}{4}} \tan^{n-2} \theta \sec^2 \theta d\theta \\ &= \left[\frac{\tan^{n-1} \theta}{n-1} \right]_0^{\frac{\pi}{4}} = \frac{1}{n-1} - 0 = \frac{1}{n-1} \end{aligned}$$

$$(iii) \text{ For } 0 < \theta < \frac{\pi}{4},$$

$$0 < \tan \theta < 1$$

(since $\tan \theta$ is monotonic increasing on the domain)

$$0 < \tan^2 \theta < 1$$

$$0 < \tan^n \theta < \tan^{n-2} \theta$$

(since $\tan^{n-2} > 0$)

$$\therefore \int_0^{\frac{\pi}{4}} \tan^n \theta d\theta < \int_0^{\frac{\pi}{4}} \tan^{n-2} \theta d\theta$$

that is, $I_n < I_{n-2}$

$$\therefore 2I_n < I_{n-2} + I_n \quad (1)$$

$$2I_n < \frac{1}{n-1} \quad [\text{from (ii)}]$$

$$I < \frac{1}{2(n-1)}$$

Replacing n by $n+2$ in (1), we have

$$I_{n+2} < I_n$$

$$I_{n+2} + I_n < 2I_n$$

$$\frac{1}{n+2-1} < 2I_n \quad [\text{from (ii)}]$$

$$\frac{1}{2(n+1)} < I_n$$

$$\text{Hence } \frac{1}{2(n+1)} < I_n < \frac{1}{2(n-1)}.$$

$$I_3 = \frac{1}{4} - I_1 \quad [\text{from (ii)}]$$

$$= \frac{1}{4} - \left(\frac{1}{2} - I_1\right) \quad [\text{from (ii)}]$$

$$= -\frac{1}{4} + \frac{1}{2} \ln 2 \quad [\text{from (i)}]$$

$$= \frac{2 \ln 2 - 1}{4}$$

From (iii), we have

$$\frac{1}{2 \times 6} < I_5 < \frac{1}{2 \times 4}$$

$$\frac{1}{12} < \frac{2 \ln 2 - 1}{4} < \frac{1}{8}$$

$$\frac{1}{3} < 2 \ln 2 - 1 < \frac{1}{2}$$

$$\frac{4}{3} < 2 \ln 2 < \frac{3}{2}$$

$$\frac{2}{3} < \ln 2 < \frac{3}{4} \quad \text{as required.}$$



Further Integration Solution

$$\begin{aligned} (i) \quad I_n + I_{n-1} &= \int_0^1 \frac{x^{2n}}{1+x^2} dx + \int_0^1 \frac{x^{2n-2}}{1+x^2} dx \\ &= \int_0^1 \left(\frac{x^{2n}}{1+x^2} + \frac{x^{2n-2}}{1+x^2} \right) dx \\ &= \int_0^1 \frac{x^{2n-2}(x^2+1)}{1+x^2} dx \\ &= \int_0^1 x^{2n-2} dx = \frac{1}{2n-1} [x^{2n-1}]_0^1 \\ &= \frac{1}{2n-1} \end{aligned}$$

$$\begin{aligned} (ii) \quad \int_0^1 \frac{x^4}{1+x^2} dx &= I_2 \\ &= \frac{1}{3} - I_1 \quad (\text{from (i)}) \\ &= \frac{1}{3} - (1 - I_0) \quad (\text{from (i)}) \\ &= -\frac{2}{3} + \int_0^1 \frac{dx}{1+x^2} \\ &= -\frac{2}{3} + [\tan^{-1} x]_0^1 \\ &= \boxed{\frac{\pi}{4} - \frac{2}{3}} \end{aligned}$$



Further Integration Solution

$$\begin{aligned} (i) \quad R.H.S. &= (1-\sqrt{x})^{n-1} - (1-\sqrt{x})^n \\ &= (1-\sqrt{x})^{n-1} [1 - (1-\sqrt{x})] \quad (1) \\ &= (1-\sqrt{x})^{n-1} [1 - 1 + \sqrt{x}] \\ &= \sqrt{x} (1-\sqrt{x})^{n-1} = L.H.S. \end{aligned}$$

$$(ii) \quad I_n = \int_0^1 (1-\sqrt{x})^n dx$$

$$\text{Let } u = (1-\sqrt{x})^n \quad dv = dx$$

$$\frac{du}{dx} = n(1-\sqrt{x})^{n-1} \times \frac{-1}{2\sqrt{x}} \quad v = x$$

$$\therefore I_n = uv - \int v du \quad (2)$$

$$= \left[x(1-\sqrt{x})^n \right]_0^1 - \int_0^1 \frac{n x (1-\sqrt{x})^{n-1}}{2\sqrt{x}} dx$$

$$= (0-0) + \frac{n}{2} \int_0^1 \sqrt{x} (1-\sqrt{x})^{n-1} dx$$

$$(\text{from part i}) \Rightarrow \frac{n}{2} \int_0^1 [(1-\sqrt{x})^{n-1} - (1-\sqrt{x})^n] dx \quad (3)$$

$$= \frac{n}{2} \int_0^1 (1-\sqrt{x})^{n-1} dx - \frac{n}{2} \int_0^1 (1-\sqrt{x})^n dx$$

$$\therefore I_n = \frac{n}{2} I_{n-1} - \frac{n}{2} I_n$$

$$I_n + \frac{n}{2} I_n = \frac{n}{2} I_{n-1}$$

$$\left(\frac{2+n}{2} \right) I_n = \frac{n}{2} I_{n-1}$$

$$\therefore I_n = \frac{n}{2} \times \frac{2}{2+n} I_{n-1} \quad (4)$$

$$I_n = \frac{n}{n+2} I_{n-1}, \text{ as required}$$

$$(iii) \quad I_n = \frac{n}{n+2} I_{n-1} \quad (n \geq 1)$$

$$I_1 = \frac{1}{1+2} I_0$$

$$= \frac{1}{3} \times \int_0^1 (1-\sqrt{x})^0 dx$$

$$= \frac{1}{3} \times [x]_0^1 = \frac{1}{3} \times 1$$

$$= \frac{1}{3} \quad (\text{That is: } I_0 = 1)$$

$$I_{2022} = \frac{2022}{2024} I_{2021}$$

$$= \frac{2022}{2024} \times \frac{2021}{2023} I_{2020}$$

$$= \frac{2022}{2024} \times \frac{2021}{2023} \times \frac{2020}{2022} I_{2019}$$

$$= \frac{2022 \times 2021 \times 2020 \times \dots \times 4 \times 3 \times 2}{2024 \times 2023 \times 2022 \times \dots \times 4 \times 3} I_1$$

$$= \frac{3 \times 2}{2024 \times 2023} \times \frac{1}{3} = \frac{1}{2 \times 047 \times 276}$$



Further Integration Solution

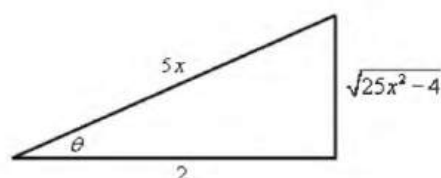
$$x = \frac{2}{5} \sec \theta \Rightarrow dx = \frac{2}{5} \sec \theta \tan \theta d\theta$$

$$\begin{aligned} \int \frac{\sqrt{25x^2-4}}{x} dx &= \int \frac{2 \tan \theta}{\frac{2}{5} \sec \theta} \left(\frac{2}{5} \sec \theta \tan \theta d\theta \right) \\ &= 2 \int \tan^2 \theta \end{aligned}$$

$$= 2 \int \sec^2 \theta - 1 d\theta$$

$$= 2(\tan \theta - \theta) + C$$

$$\text{Since } \sec \theta = \frac{5x}{2} = \frac{\text{hyp}}{\text{adj}}$$



$$\therefore \tan \theta = \frac{\sqrt{25x^2-4}}{2}$$

$$\& \cos \theta = \frac{2}{5x} \Rightarrow \theta = \cos^{-1} \left(\frac{2}{5x} \right)$$

$$\therefore \int \frac{\sqrt{25x^2-4}}{x} dx = \sqrt{25x^2-4} - 2 \cos^{-1} \left(\frac{2}{5x} \right) + C$$

